

REPORT ON WATER QUALITY INDICATORS FOR COMMUNITY-BASED MONITORING PLATFORM

ACADEMIC DEFENSE PROJECT – BAMENDA, CAMEROON

Prepared for: Academic Supervisor

Subject: Research Findings on Water Quality Indicators and Proposed Monitoring Framework

1. Executive Summary

This report presents research findings from online literature review and field consultations regarding water quality indicators suitable for a community-based software monitoring platform. The investigation focused on identifying measurable parameters that can be effectively collected by diverse actors—community members, water workers, and health officials—to provide actionable insights for water quality management in Bamenda, Cameroon.

2. Research Methodology

2.1 Online Sources Consulted

- World Health Organization (WHO) Guidelines for Drinking-water Quality: Small Water Supplies (2024)
- Peer-reviewed studies on water quality in Bamenda, Cameroon
- Water Quality Index development research for African contexts
- Regional case studies from Nigeria and Ghana
- International water quality testing parameters

2.2 Field Consultations

- Water maintenance officer at Akom drinking water dam
- Community members in Bamenda regarding water security issues

3. Proposed Water Quality Indicators

Based on international standards and local feasibility, indicators are organized by who can collect them and their public health significance.

3.1 Physical Indicators (Community-Level Reporting)

These indicators can be reported by community members without specialized equipment:

Indicator	Description	Reporting Method	Health Significance
Turbidity	Water clarity/clarity	Visual: "Clear," "Cloudy," "Very muddy"	High turbidity can harbor pathogens and reduce disinfection effectiveness
Colour	Unusual water coloration	Visual observation	Indicates organic matter, algae, or industrial contamination
Odor	Unusual smell	Sensory: "Normal," "Sewage smell," "Chemical smell"	Often indicates bacterial activity or chemical pollution
Visible Contamination	Floating debris, oil sheen, dead animals	Direct observation	Immediate health risk indicator

3.2 Chemical Indicators (Water Worker-Level Testing)

These require basic field testing kits that water workers already possess or can be trained to use:

Indicator	Target/Range	Testing Method	Why It Matters
pH	6.5–8.5	Test strips or meter	Affects disinfection; extreme pH causes health issues and pipe corrosion
Electrical Conductivity (EC)	< 2,500 $\mu\text{S}/\text{cm}$	Conductivity meter	Measures dissolved solids; high levels indicate pollution
Dissolved Oxygen (DO)	> 5 mg/L	DO meter or kit	Low DO indicates organic pollution and poor water quality
Temperature	Ambient (recorded)	Thermometer	Affects biological activity and disinfection

3.3 Biological Indicators (Health Official Reporting)

These are collected by health officials tracking disease outbreaks:

Indicator	What It Detects	Reporting Mechanism
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E. coli / Fecal Coliforms	Fecal contamination	Laboratory results or rapid test kits
Reported Diarrhea Cases	Waterborne disease incidence	Health facility records
Cholera/Typhoid Outbreaks	Serious waterborne diseases	Official outbreak reports
Staphylococcus, Salmonella, Shigella	Pathogen presence	Clinical laboratory data

3.4 Composite Indicator: Water Quality Index (WQI)

Research from African contexts indicates that **Water Quality Indices (WQIs)** are valuable tools for combining multiple parameters into single, understandable values. The two most commonly adapted WQIs in Africa are:

1. **Weighted Arithmetic WQI (WAWQI)** – Suitable for drinking water assessment
2. **Canadian Council of Ministers of Environment WQI (CCMEWQI)** – Suitable for surface water monitoring

5. Locations for Data Collection

Based on existing research and local knowledge, the following locations should be prioritized for monitoring:

4.1 Primary Water Sources in Bamenda

Location/Zone	Source Type	Priority Level	Rationale
Akom Drinking Water Dam	Surface water/Dam	HIGH	Your initial research site; serves significant population
Pipe-borne water distribution points	Municipal supply	HIGH	Primary water source for 65% of households
Boreholes across three Sub-Divisions	Groundwater	MEDIUM	Secondary source for 18.6% of population
Springs and streams	Natural sources	MEDIUM	Used by 8.1% of population; often unprotected
Household wells	Groundwater	LOW	Used by 8.3% of population; high contamination risk

4.2 Geographic Coverage

The three Sub-Divisions of Bamenda to be covered:

1. Bamenda I
2. Bamenda II
3. Bamenda III

5. Stakeholder Roles in Data Collection

Based on the WHO framework for small water supplies, multiple stakeholders contribute to water safety:

Stakeholder	Role	Data to Provide	Frequency
Water maintenance officers	Professional testing	pH, EC, DO, temperature, turbidity	Daily/Weekly
Community members	Local observation	Visual quality, odor, color, pipe leaks	Real-time
Health officials	Disease surveillance	Diarrhea cases, outbreak reports, lab results	Weekly/Monthly
NGOs	Response coordination	Intervention records, repair status	As needed
Ministry of Water Resources	Policy oversight	Regulatory data, infrastructure updates	Monthly

6. How This Platform Addresses Gaps

6.1 No Existing Platform in Cameroon

Research confirms that no centralized water quality data platform currently exists in Cameroon. Your proposed system would be a **first-of-its-kind** initiative.

6.2 Moving Beyond Hardware Limitations

Hardware-Only Approach	Your Software Platform
Measures only chemical parameters	Integrates human observation + sensor data
No coordination mechanism	Direct NGO/technician alerts

Expensive to scale	Scales via existing mobile phones
Requires spare parts supply chain	Remotely updatable
Single point of failure	Distributed reporting network

6.3 Data-to-Action Pipeline

This platform enables:

Community Report → Dashboard Alert → NGO Notification → Repair Dispatch → Follow-up Confirmation

7. Technical Recommendations

7.1 Data Validation

Implement basic validation rules:

- pH must be between 0-14
- Temperature must be reasonable for tropical climate (15-40°C)
- Geographic coordinates required for each report

8. Conclusion

The research confirms that a software-based monitoring platform is both **feasible** and **urgently needed** in Bamenda. The WHO 2024 Guidelines explicitly support risk management approaches that combine multiple data sources—exactly what your platform provides.

The proposed indicators are:

- **Scientifically valid** (based on WHO and peer-reviewed research)
- **Locally appropriate** (informed by Bamenda-specific studies)
- **Practically collectible** (matched to stakeholder capabilities)
- **Actionable** (leading to specific interventions)

9. Request for Approval

Based on the evidence presented in this report, I respectfully request your approval to proceed with the software-based monitoring platform approach, with the indicator framework outlined above serving as the foundation for system design.

Next steps upon approval:

1. Finalize indicator definitions and thresholds
2. Design data collection forms for each stakeholder type
3. Develop prototype dashboard
4. Pilot testing at Akom drinking water dam

References

1. WHO. (2024). Guidelines for drinking-water quality: small water supplies. Geneva: World Health Organization.
2. African Water Association. (2025). State of Urban Household Drinking Water Security Situation in Bamenda North West Region, Cameroon.
3. African Water Association. (2025). Effectiveness and health risk assessment of drinking water from different sources treated by local household water treatment methods in Bamenda, Cameroon.
4. ACECoR. (2025). Adapted Water Quality Indices: Limitations and Potential for Water Quality Monitoring in Africa.
5. Missouri Department of Natural Resources. Water Quality Testing Parameters.